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221754

**5th Sem/
Branch : Mechanical Engg.
Subject : Theory of Machines**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 A simple mechanism has _____

- | | |
|------------|------------|
| a) 4 links | b) 1 links |
| c) 3 links | d) 2 links |

Q.2 A ball and a socket forms a

- | | |
|-----------------|-------------------|
| a) Turning pair | b) Screw pair |
| c) Rolling Pair | d) Spherical pair |

Q.3 Which one of the following is a Dead weight type governor?

- a) Porter Governor
- b) Wilson-Hartnell Governor
- c) Hartnell Governor
- d) Hartung Governor

- Q.4 In a cam-follower mechanism, the follower motion is controlled by the :
- a) Shape of the follower
 - b) Speed of cam
 - c) Shape of cam
 - d) Mass of the follower
- Q.5 The gear train usually employed in clocks is:
- a) Reverted gear train
 - b) Simple gear train
 - c) Sun & planet gear train
 - d) Differential gear train
- Q.6 In balancing of single-cylinder engine, the rotating unbalance is _____
- a) Completely made zero and so also the reciprocating unbalance
 - b) Completely made zero and the reciprocating unbalance is partially reduced
 - c) Partially reduced and the reciprocating unbalance is completely made zero
 - d) Partially reduced and so also the reciprocating unbalance

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 What is initial tension in belt?
- Q.8 Define dynamic balancing.
- Q.9 Define hunting in governor.
- Q.10 What is crowning in pulley?
- Q.11 What is diametral pitch?
- Q.12 What is slip in a belt drive?

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Classify chain drive.
- Q.14 Explain the working of porter governor.
- Q.15 Differentiate between fly wheel and governor.
- Q.16 Explain compound gear train.
- Q.17 Explain fast and loose pulley.
- Q.18 Draw the turning moment diagram for single acting and double acting steam engine.
- Q.19 Obtain an expression of ratio of belts in flat belt drive.
- Q.20 Explain the terminology of cam profile.
- Q.21 Differentiate between machine and mechanism.
- Q.22 Explain the importance of balancing in machines.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain single slider crank chain mechanisms in detail. Explain its inversions.
- Q.24 What are causes of vibration in machines? What are their remedial measures? Also explain their harmful effects.
- Q.25 A cam is to give the following motion to a knife edged follower:
- Out stroke 60° of cam rotation with uniform velocity
 - Dwell for 30° .
 - Return stroke 60° with uniform velocity.
 - Dwell for remaining 210° . Stroke of follower is 40mm and minimum radius of cam is 50 mm. Draw the cam profile, when the axis of follower passes through the axis of cam shaft.